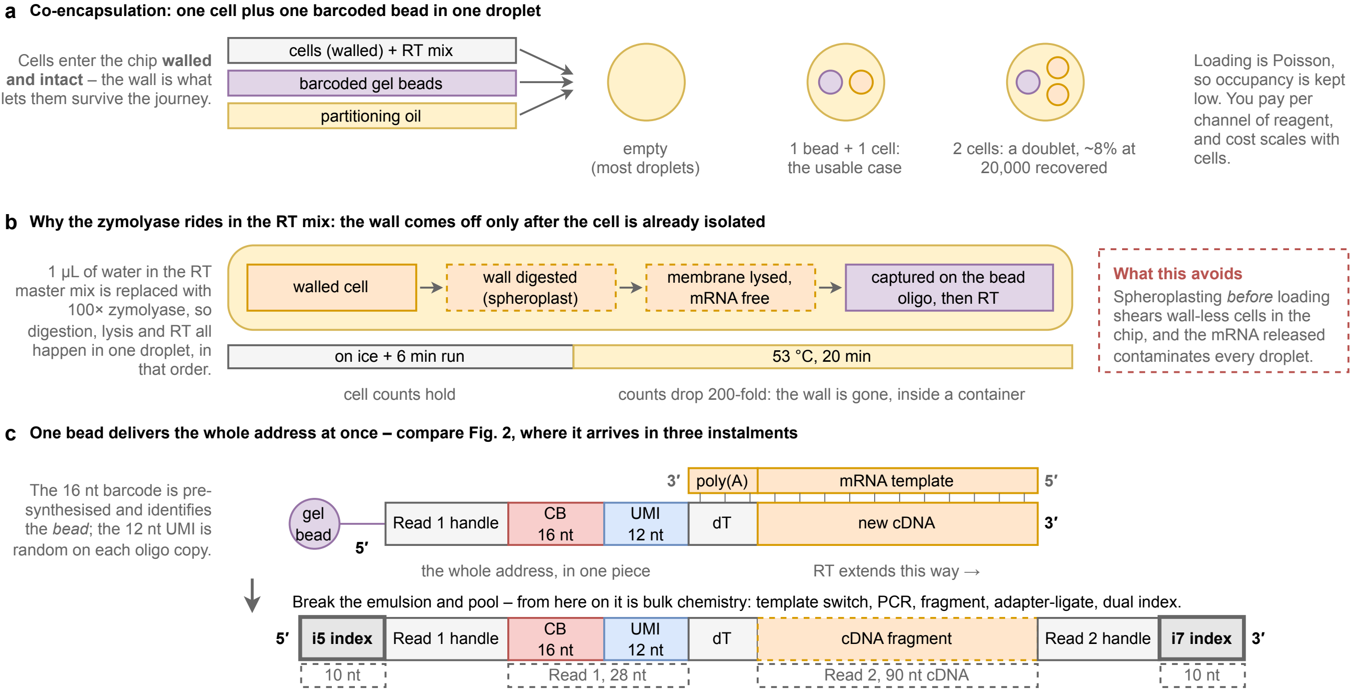
Droplet barcoding (10x Chromium): cell identity is bought with a container, not built into the sequence

Yeast-adapted 10x Chromium of Jariani 2020 Panels read a to c: the container is made, the wall comes off inside it, and a barcoded molecule comes out. Segment colors and the 5′-to-3′ convention are those of Fig. 2, so the two methods can be compared directly.



Both families put the barcode on the reverse-transcription primer; what differs is the thing that holds one cell's molecules together. Split-pool uses the fixed cell as the vessel and writes the address over three ligation rounds (Fig. 2); droplet rents an oil-walled container and delivers a 16 nt bead barcode plus a 12 nt UMI in one step, which is why its run needs 118 cycles against 160. The container is also the constraint: it must exist before the wall comes off, and it is paid for per channel.

Segments transcript / cDNA | cell barcode | UMI | gel bead | droplet / oil | handle / adapter | Illumina index | ‖ base pairing | dashed = disrupted